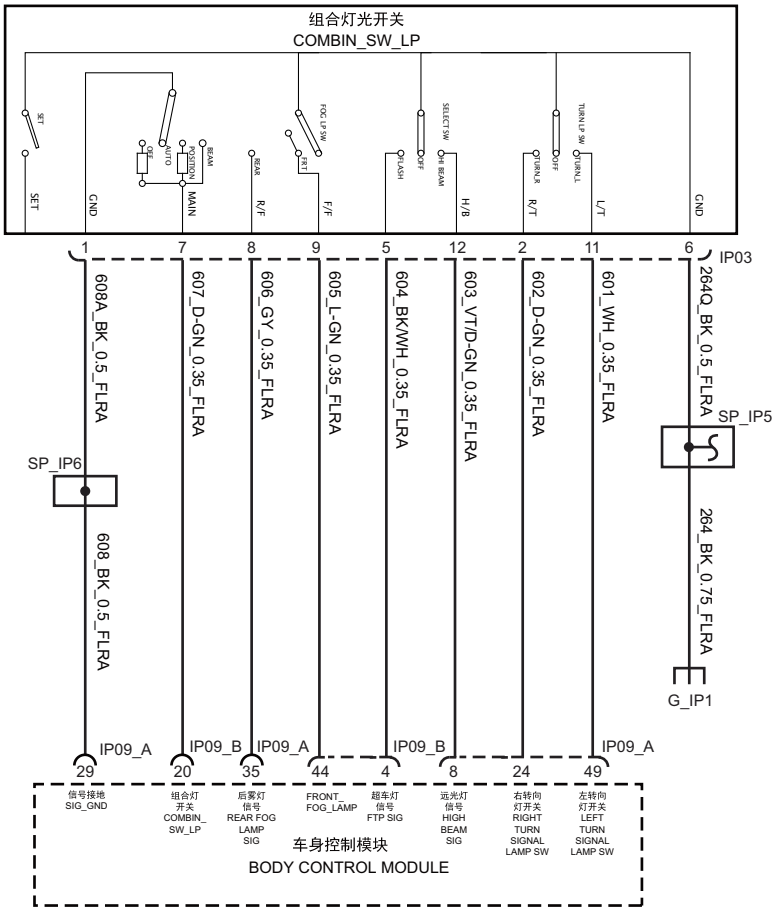
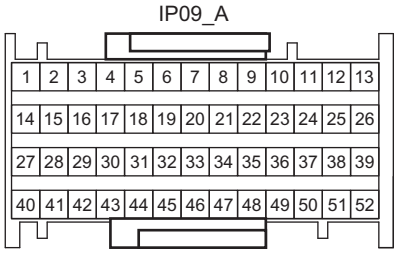
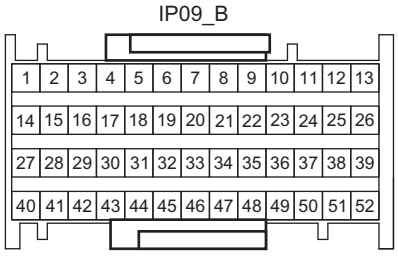
B1201-11、 B1201-12、 B1201-13、 B1201-1E

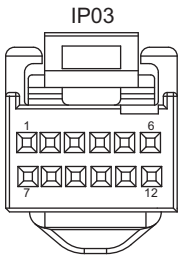
Fault diagnosis code
B1201-11：Combination lights switch circuit short to ground
B1201-12：Combination lights switch circuit short to battery
B1201-13：Combination lights switch circuit open
B1201-1E：Combination lights switch circuit resistance out of range
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • BCM fault • Insurance fuse fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
B1201-11
Combination lights switch circuit short to ground
B1201-12
Combination lights switch circuit short to battery
B1201-13
Combination lights switch circuit open
B1201-1E
The input resistance of combination light switch is detected to be out of range for more than 1s.
To set the effect of a fault code condition
1 Combination lights switch circuit fails 2 Combination lights switch circuit function fails.
Description of circuit diagram

Circuit diagram

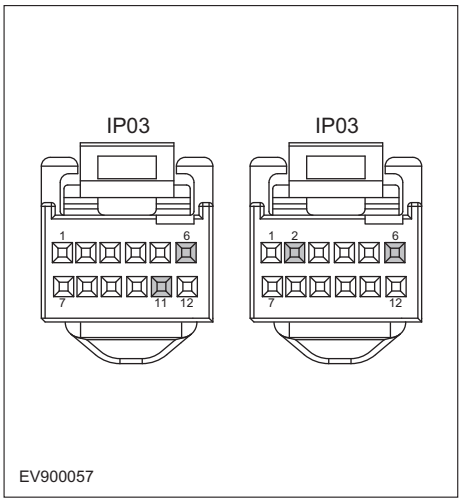


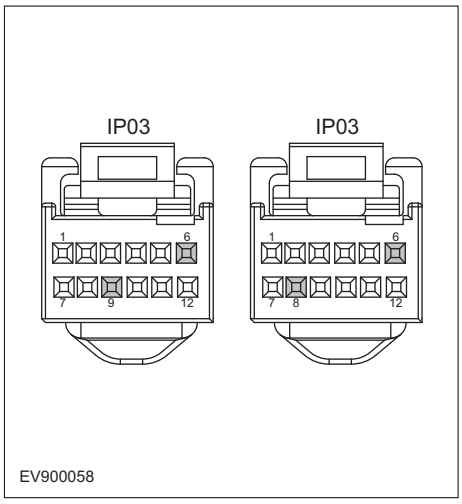
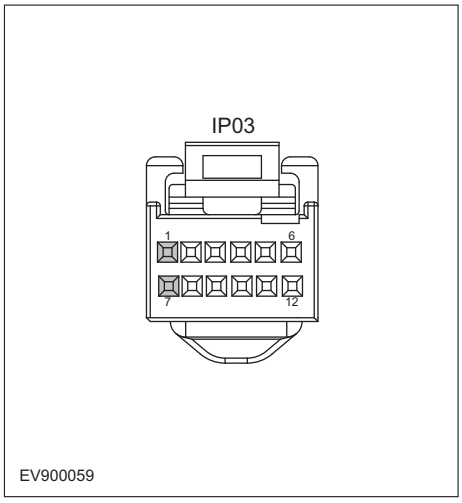




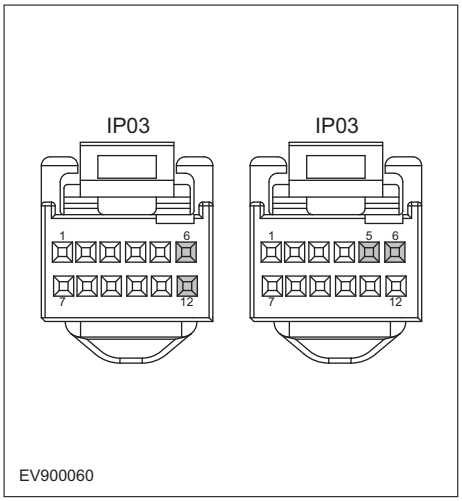


EV900056

<table>
<tr><td colspan="3">Troubleshooting steps</td></tr>
<tr><td>Step 1</td><td colspan="2">General inspection.</td></tr>
<tr><td colspan="2"></td><td>1 Check the fuse for damage.
2 Check battery capacity.
3 Check the harness plug of air conditioning control module for damage, poor contact, aging and looseness.
4 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO ➞ Repair or replace the faulted parts.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
<tr><td>Step 2</td><td colspan="2">Check fault codes.</td></tr>
<tr><td colspan="2"></td><td>1 Connect the fault diagnosis instrument to the diagnosis interface.
2 Turn the ignition switch to ON position.
3 Check the body control module with a diagnostic apparatus.
4 Is there any other fault code except for B1201-11、B1201-12、 B1201-13、 B1201-1E?</td></tr>
<tr><td colspan="2"></td><td>YES ➞ Refer to:DTC Summary list(BCM).</td></tr>
<tr><td colspan="3">NO ↓</td></tr>
<tr><td>Step 3</td><td colspan="2">Check the combination lights switch (turning light switch)(parts).</td></tr>
<tr><td colspan="2"> 
</td><td>1 Turn the ignition switch to OFF.
2 Disconnect the harness plug IP03 of IPK module.
3 Place the combination light switch in the “left turn signal”position and measure the resistance between the combination light switch terminals(combination switch).
Measuring circuit: resistance between terminal 11 on plug IP03 and terminal 6 on plug IP03.
Standard value: ≈ 0Ω
4 Place the combination light switch in the “right turn signal”position and measure the resistance between the combination light switch terminals(combination switch).
Measuring circuit: resistance between terminal 2 on plug IP03 and terminal 6 on plug IP03.
Standard value: ≈ 0Ω
5 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO ➞ Replace combination lights.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
</table>

Step 4	Check the combination lights switch (fog lamp switch)(parts).
	1 Turn the ignition switch to OFF. 2 Place the combination light switch in the “front fog lamp”position and measure the resistance between the combination light switch terminals(combination switch). Measuring circuit: resistance between terminal 9 on plug IP03 and terminal 6 on plug IP03. **Standard value: ≈ 0Ω** 3 Place the combination light switch in the “rear fog lamp”position and measure the resistance between the combination light switch terminals(combination switch). Measuring circuit: resistance between terminal 8 on plug IP03 and terminal 6 on plug IP03. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Replace combination lights.
YES ↓	
Step 5	Check the combination lights switch (low beam light switch)(parts).
	1 Turn the ignition switch to OFF. 2 Place the combination light switch in the “low beam light”position and measure the resistance between the combination light switch terminals(combination switch). Measuring circuit: resistance between terminal 7 on plug IP03 and terminal 1 on plug IP03. **Standard value: ≈ 0Ω** 3 Check whether the result is normal or not?
	NO → Replace combination lights.
YES ↓	

Step 6	Check the combination lights switch (high beam light switch)(components).



1 Turn the ignition switch to OFF.

2 Place the combination light switch in the “high beam light”position and measure the resistance between the combination light switch terminals(combination switch).

Measuring circuit: resistance between terminal 12 on plug IP03 and terminal 6 on plug IP03.

Standard value: ≈ 0Ω

3 Place the combination light switch in the “flashlight”position and measure the resistance between the combination light switch terminals (combination switch).

Measuring circuit: resistance between terminal 5 on plug IP03 and terminal 6 on plug IP03.

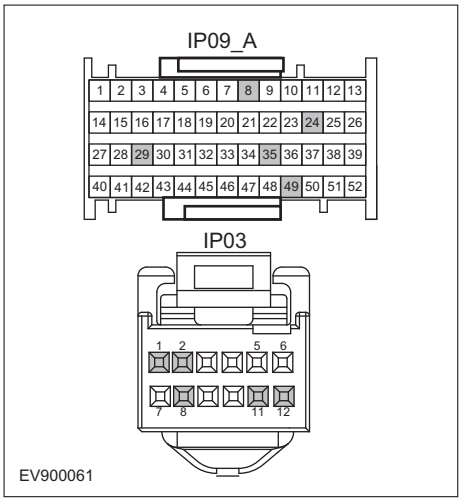
Standard value: ≈ 0Ω

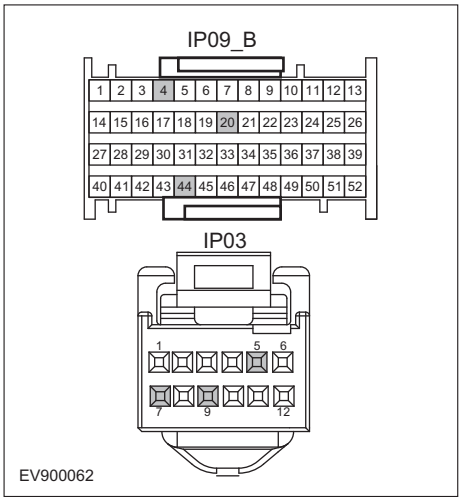
4 Check whether the result is normal or not?

NO → Replace combination lights.

YES ↓

Step 7	Check the circuit of combination light switch and BCM (open circuit).





1 Turn the ignition switch to OFF.

2 Disconnect the harness plug IP09_A, IP09_B of BCM module.

3 Disconnect the harness plug IP03 of IPK module.

4 Measure the resistance between the harness terminal of combination light switch and the BCM.

Measuring circuit: resistance between terminal 1 on plug IP03 and terminal 29 on plug IP09_A.

Measuring circuit: resistance between terminal 2 on plug IP03 and terminal 24 on plug IP09_A.

Measuring circuit: resistance between terminal 8 on plug IP03 and terminal 35 on plug IP09_A.

Measuring circuit: resistance between terminal 11 on plug IP03 and terminal 49 on plug IP09_A.

Measuring circuit: resistance between terminal 12 on plug IP03 and terminal 8 on plug IP09_A.

Measuring circuit: resistance between terminal 5 on plug IP03 and terminal 4 on plug IP09_B.

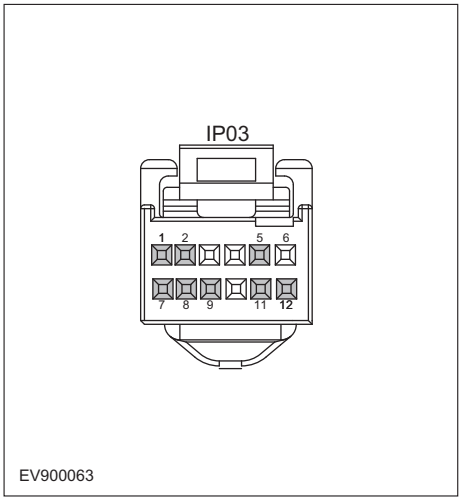
Measuring circuit: resistance between terminal 7 on plug IP03 and terminal 20 on plug IP09_B.

Measuring circuit: resistance between terminal 9 on plug IP03 and terminal 44 on plug IP09_B.

Standard value: ≈ 0Ω

5 Check whether the result is normal or not?

	NO →	Repair or replace the faulted parts.
YES ↓		
Step 8	Check the circuit of combination light switch (short circuit).	



1 Turn the ignition switch to OFF.

2 Connect the harness plug IP09_A, IP09_B of body control module.

3 Measure the resistance between harness terminals on combination switch light and ground.

Measuring circuit: resistance between terminal 1 on plug IP03 and ground.

Measuring circuit: resistance between terminal 2 on plug IP03 and ground.

Measuring circuit: resistance between terminal 5 on plug IP03 and ground.

Measuring circuit: resistance between terminal 7 on plug IP03 and ground.

Measuring circuit: resistance between terminal 8 on plug IP03 and ground.

Measuring circuit: resistance between terminal 9 on plug IP03 and ground.

Measuring circuit: resistance between terminal 11 on plug IP03 and ground.

Measuring circuit: resistance between terminal 12 on plug IP03 and ground.

Standard value: ∞

4 Measure the voltage between harness terminals on combination light switch and ground.

Measuring circuit: voltage between terminal 1 on plug IP03 and ground.

Measuring circuit: voltage between terminal 2 on plug IP03 and ground.

Measuring circuit: voltage between terminal 5 on plug IP03 and ground.

Measuring circuit: voltage between terminal 7 on plug IP03 and ground.

Measuring circuit: voltage between terminal 8 on plug IP03 and ground.

Measuring circuit: voltage between terminal 9 on plug IP03 and ground.

Measuring circuit: voltage between terminal 11 on plug IP03 and ground.

Measuring circuit: voltage between terminal 12 on plug IP03 and ground.

Standard value: ≈ 0V

5 Check whether the result is normal or not?

	NO → Repair or replace the faulted parts.

YES ↓		
Step 9	Test whether BCM is norm.	
		1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB09_C、EB09_D、IP09_A、IP09_B of BCM. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BCM plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
		YES → Replace the BCM.
NO ↓		
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,		
Next Step ↓		
Diagnosis end.		